

# Inference of Protein Secondary Structure with HMMs and BayesFlow

Simulation-Based Inference — Project Report

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# Introduction & Modeling Idea

## The Problem

Predict per-position secondary structure  
from variable-length amino acid sequences.

## Two-State HMM

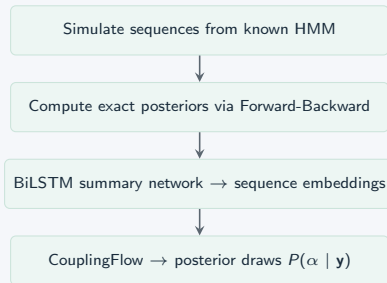
State 0 =  $\alpha$ -helix

State 1 = other ( $\beta$ -sheets & coils)

## Modeling Goal

Approximate  $P(S_t = \alpha \mid \mathbf{y})$  at every position  $t$   
using **BayesFlow** neural posterior estimation.

## Overall Pipeline



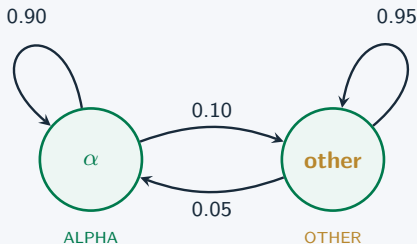
## Validation

Compare to experimental ground truth:

Human insulin **1A7F** (chain A: 21 aa, chain B: 29 aa).

# The 2-State Hidden Markov Model

## State Transitions



**Deterministic start:**  $P(S_1 = \text{other}) = 1.0$ ,  $P(S_1 = \alpha) = 0.0$

**Highly persistent:**  $\alpha$  stays  $\alpha$  (90%), other stays other (95%)

## Selected Emission Contrasts

| Amino acid   | $\alpha$ -helix | other | Role          |
|--------------|-----------------|-------|---------------|
| A (Alanine)  | 12%             | 6%    | Strong former |
| E (Glutamic) | 9%              | 5%    | Former        |
| P (Proline)  | 2%              | 6%    | Breaker       |
| G (Glycine)  | 4%              | 9%    | Breaker       |

Full  $20 \times 2$  table in written report.

## Prior Dynamics

**Stationary distribution:**

$$\pi(\alpha) = 0.05 / (0.05 + 0.10) = 1/3 \approx 33\%$$

$$\pi(\text{other}) \approx 67\%$$

Prior strongly favors **non-helical** states ( $\sim 67\%$ ).

# Data Generation & Analytical Targets

## Simulator

Sequential Markov sampling of hidden states.

Independent randomized emission draws.

Random sequence lengths: 10–60 aa.

**100% simulated** — no real proteins.

## Target Computation

Exact posteriors via Forward-Backward.

`hmmlearn.CategoricalHMM` — manually

configured, never `.fit()`.

## Verification

Targets verified against independent

pure-NumPy FB solver. Confirms:

$P(S_0 = \alpha) \equiv 0$  (position 0 = seq. position 1).

## On-demand Generation

`generate_dataset(n, ...)` produces

any number of (sequence, posterior, states) tuples. Downstream code

chooses the split sizes.

## Variable-Length by Design

Sequences kept **ragged** (10–60 aa),  
each stored with its own length.

The data layer does **not pad** — padding  
& masking are handled downstream by

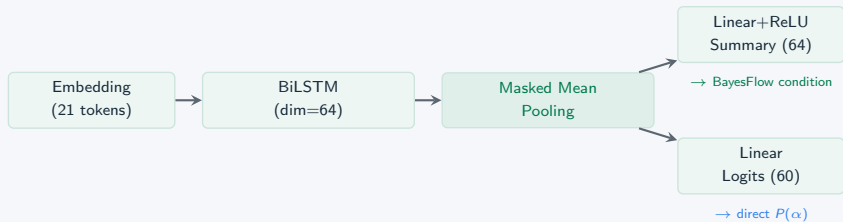
the summary network.

## Storage

`.npz` format: concatenated arrays +

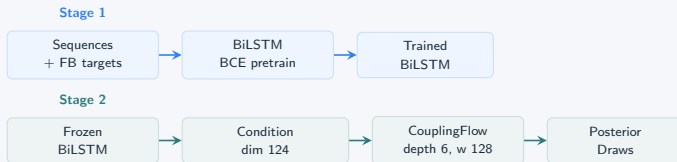
`lengths` vector. No object arrays.

# The Summary Network — BiLSTM



1. BiLSTM captures **sequential** dependencies — permutation-invariant nets fail here.
2. Padding token 20 & masked loss → handles variable-length sequences.
3. Two-stage: BiLSTM pretrained on BCE → frozen → CouplingFlow on top.

# Two-Stage Training & CouplingFlow



1. BiLSTM pretrained on masked BCE → frozen → condition (dim 124).
2. CouplingFlow: depth 6, hidden 128, affine, actnorm → posterior draws.
3. **Model selection:** **not** flow val\_loss. Rank by masked MAE & correlation.

# Diagnostics: Adapted Simulation-Based Calibration

## Standard SBC

Draw  $\theta \sim p(\theta)$  from prior.

Simulate data  $\mathbf{x} \sim p(\mathbf{x} \mid \theta)$ .

Compute posterior, check if ranks of  $\theta_{\text{true}}$  are uniform.

## Our Challenge

HMM parameters are **fixed & known**.

$P(\alpha_t \mid \mathbf{y})$  is a **deterministic** function of the input — no parameter uncertainty to recover.

## Our Adapted Method

1. Draw  $S$  test sequences from the HMM.



2. Generate  $M$  posterior draws from CouplingFlow at each position  $t$ .



3. Rank the true FB target value within the flow's posterior draws.



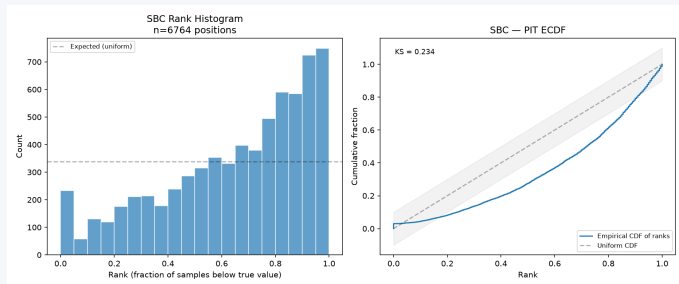
4. Test uniformity: rank histogram, PIT ECDF, KS statistic.

## What This Tests

Does the BayesFlow posterior **envelope** the deterministic FB value with correct frequency?

Note: Not a classical prior-predictive SBC — adapted for the deterministic-target setting.

# Calibration Results — Adapted SBC



## Systematic Underestimation

Rank histogram shifted **rightward**:

mean rank = **0.65**

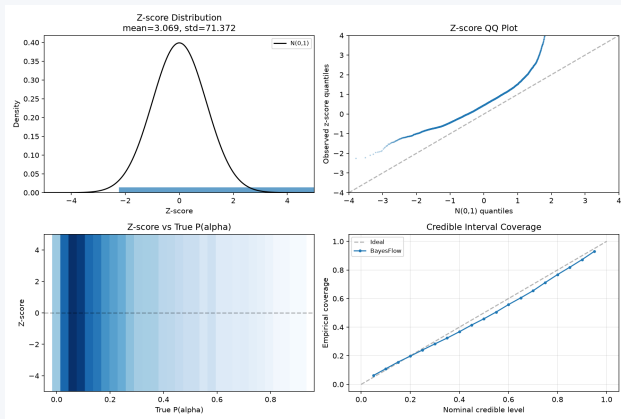
only **27%** below 0.50

**KS = 0.234**

True FB  $P(\alpha)$  is frequently **larger** than posterior draws.

→ Padded zero boundaries pull the continuous flow toward 0.

# Calibration Results — Z-Score & Coverage



## Undercoverage & Bias

Raw z-score std = **71.4**

Trimmed (1.1% outliers removed):  
mean = **0.63**, std = **1.27**

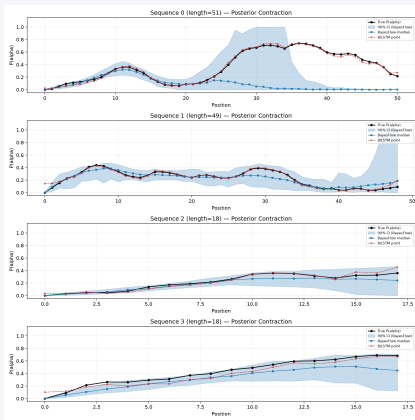
87% empirical vs. 90% nominal

93% empirical vs. 95% nominal

Posteriors are **too narrow**  
and **slightly biased**.

→ Localized overconfidence  
at sequence boundaries.

# Synthetic Evaluation — Posterior Contraction



Black = true  $P(\alpha)$ . Blue band = 90% CI. Red = BiLSTM point estimate.

## BiLSTM Wins on Accuracy

|      | BayesFlow | BiLSTM       |
|------|-----------|--------------|
| MAE  | 0.115     | <b>0.021</b> |
| Corr | 0.560     | <b>0.988</b> |

FB target is **deterministic**  
— direct network learns it  
more efficiently than a  
density estimator.

**But:** BayesFlow gives  
**uncertainty quantification**  
— credible intervals,  
SBC, z-scores, coverage.

# Real-World Validation — Human Insulin (1A7F)

## Chain B

BayesFlow

$r = 0.86$

34.5% true helix

$\approx \pi = 33\%$

## Chain A

BayesFlow

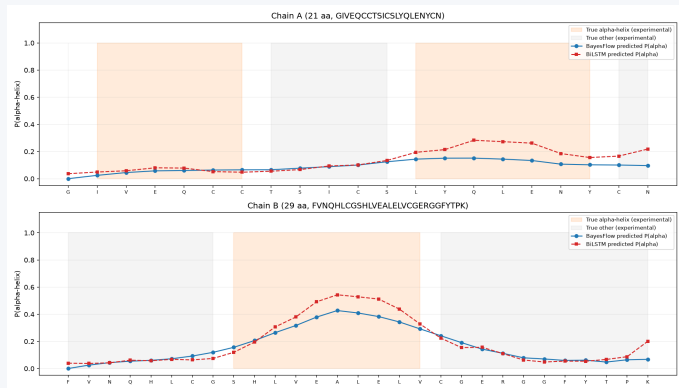
$r = 0.17$

61.9% true helix

$\gg \pi = 33\%$

**Key proof:** FB itself  
fails on Chain A  
( $r \approx 0.40$ ).

→ **Model limitation,**  
**not network failure.**



Orange = experimental  $\alpha$ -helix. Blue = BayesFlow. Red = BiLSTM.

# Limitations, Fixes & Take-Home Messages

## Limitation & Fix: The HMM

Expand to **3 states** ( $\alpha$ ,  $\beta$ , coil).

Treat transition probabilities as **inferable** parameters  $\theta$  under Beta priors.

→ BayesFlow would then recover a genuine posterior over  $\theta$ .

## Fix: The CouplingFlow

Pass targets through a **logit transformation**

before training → strictly bound all draws in  $(0, 1)$ .

## Take-Home Messages (TL;DR)

1. BiLSTM summary nets capture sequential dependencies where permutation-invariant nets fail.
2. BayesFlow successfully amortizes posterior estimation, enabling robust uncertainty quantification.
3. Real-world validation demonstrates the necessity of prior typicality checking — identify prior-data conflicts.

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Code:

[github.com/armin2080/](https://github.com/armin2080/)

[protein-secondary-structure-npe](#)